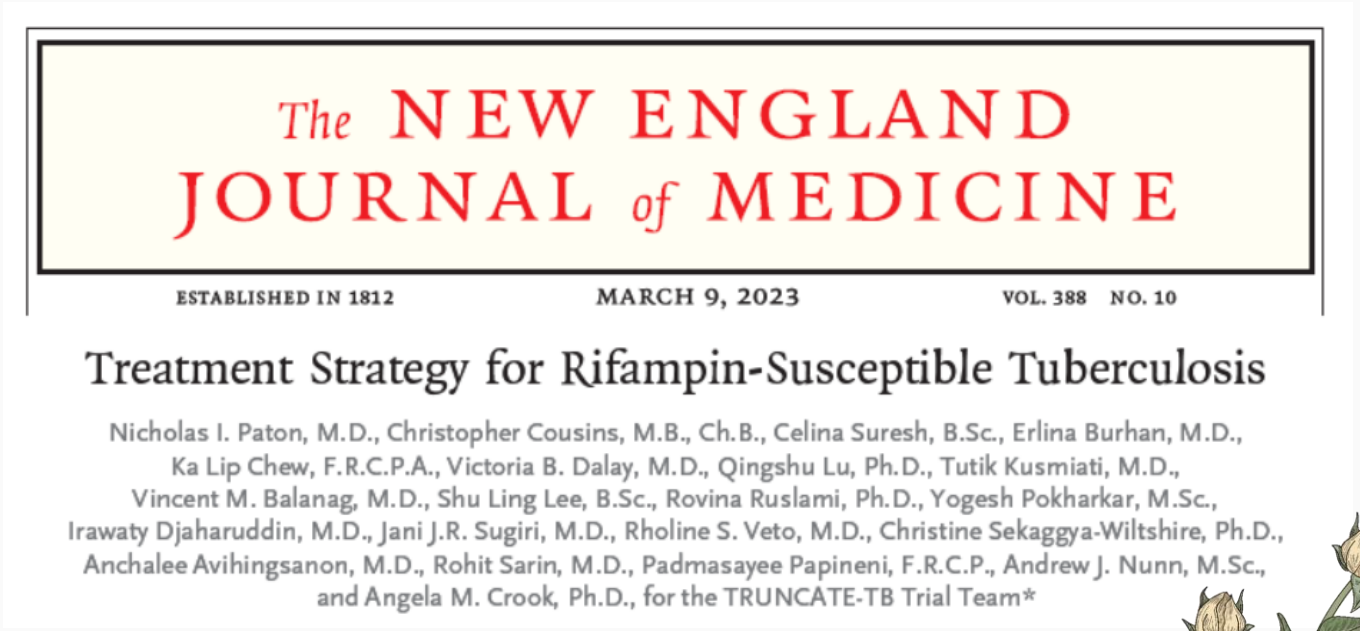
JOURNAL CLUB

Treatment Strategy for Rifampin-Susceptible Tuberculosis

Presented by Kachathan Malingam



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Treatment Strategy for Rifampin-Susceptible Tuberculosis

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BACKGROUND

Introduction

- For more than four decades, the global standard treatment for drug-susceptible pulmonary tuberculosis has been a 6-month rifampin-based regimen.
- Long-term adherence is difficult for some persons and resource constraints limit the provision of adherence support.
- The unsatisfactory outcomes associated with standard treatment have contributed to the ongoing failure to meet global tuberculosis targets and to the generation of drug resistance



5.2.2 สูตรยารักษาวัณโรค

- แนะนำสูตรยาสำหรับผู้ป่วยใหม่ที่เชื้อไวต่อยา (new patient regimen with drug susceptible) ที่ยังไม่เคยรักษา หรือเคยรักษามาไม่เกิน 1 เดือน ด้วยสูตรยา 2HRZE / 4HR
- First-line drugs (FLD) หมายถึง ยาที่ใช้สำหรับรักษาวัณโรคแนวที่ 1 ได้แก่ isoniazid, rifampicin, pyrazinamide และ ethambutol ซึ่งสูตรยาแนวที่ 1 ใช้สำหรับผู้ป่วยวัณโรครายใหม่ที่ไม่เคยรับการรักษามาก่อนหรือรับการรักษามาไม่เกิน 1 เดือนและไม่พบผลเชื้อวัณโรคดื้อต่อยา

ตารางที่ 6.1 การแบ่งกลุ่มยาสำหรับใช้ในสูตรวัณโรคดื้อยา

กลุ่มยา	ยา	
กลุ่ม A	Levofloxacin OR Moxifloxacin	Lfx Mfx
	Bedaquiline	Bdq
	Linezolid	Lzd
กลุ่ม B	Clofazimine	Cfz
	Cycloserine OR Terizidone	Cs Trd
กลุ่ม C	Ethambutol	E
	Delamanid	Dlm
	Pyrazinamide	Z
	Imipenem-cilastatin OR Meropenem	Ipm-Cln Mpm
	Amikacin (OR Streptomycin)	Am (S)
	Ethionamide OR Prothionamide	Eto Pto
	p-aminosalicylic acid	PAS

Objective

- In clinical trials, at least 85% of participants have been cured with 3-month and 4-month regimens, and the percentage is likely to be higher when these regimens contain fluoroquinolones or rifapentine.
- This study hypothesized that a strategy involving initial treatment with an 8-week regimen, extended treatment for persistent clinical disease, follow up after treatment, and prompt retreatment for the minority of persons who have a relapse might lead to long-term efficacy that would be noninferior to that of standard treatment, along with a reduced total duration of treatment.

02

METHOD

STUDY DESIGN

- Phase 2-3, international, adaptive, randomized, open-label, noninferiority trial
- Two-Month Regimens Using Novel Combinations to Augment Treatment Effectiveness for Drug-Sensitive Tuberculosis (TRUNCATE-TB) trial
- Participants were randomly assigned to undergo either standard treatment (standard dose of rifampin and isoniazid for 24 weeks in combination with pyrazinamide and ethambutol) or a strategy involving initial treatment with an 8-week regimen (used high dose of rifampin-linezolid, rifampin - clofazimine, rifapentine - linezolid, and bedaquiline - linezolid, each in combination with isoniazid, pyrazinamide and ethambutol.)





Inclusion criteria

1. Age 18 to 65 years
2. Clinical symptoms consistent with pulmonary TB and/or evidence of pulmonary TB on CXR
3. Sputum GeneXpert test positive
4. Willing to comply with the study visits and procedures
5. Resident at a fixed address that is readily accessible for visiting, within feasible travelling distance to the site and likely to remain resident there for the duration of trial follow-up
6. Willing to have directly observed therapy
7. Willing and able to provide written informed consent





Exclusion criteria

1. Taken more than 10 daily doses of standard anti-TB medication or fluoroquinolones during the 3 months prior to randomization.
2. Previous active TB disease for which treatment was given prior to the current episode
3. Severe clinical pulmonary TB e.g. respiratory failure or complications likely to require hospital admission
4. Sputum smear 3+ on sample taken at screening (using WHO/IUATLD grading system)
5. Presence of rifampicin resistance on GeneXpert test
6. Poorly-controlled diabetes that, in the opinion of the investigator, is unlikely to be controlled with available management strategies
7. Active malignancy requiring systemic chemotherapy or radiotherapy
8. Known Hepatitis B surface antigen positive and/or HCV antibody positive, unless liver function tests consistently within normal range for at least 2 years





INTERVENTION

Randomized to receive either

standard treatment	**n = 181**
rifampin–linezolid	**n = 184**
bedaquiline–linezolid	**n = 189**
rifampin–clofazimine	**n = 78**
rifapentine–linezolid	**n = 42**
N	**674**

Follow up

- Clinic visits every 1 to 4 weeks through week 24
- then every 12 weeks through week 96
- at week 30,starting monthly telephone visits were interspersed between clinic visits

A. Standard treatment arm

For first 8 weeks

DRUG	<40KG	40-54 KG	55-70KG	≥71KG
Rifampicin	300mg	450mg	600mg	750mg
Isoniazid	150mg	225mg	300mg	375mg
Pyrazinamide	800mg	1200mg	1600mg	2000mg
Ethambutol	550mg	825mg	1100mg	1375mg

For subsequent weeks to week 24

DRUG	<40KG	40-54 KG	55-70KG	≥71KG
Rifampicin	300mg	450mg	600mg	750mg
Isoniazid	150mg	225mg	300mg	375mg

B. Rifampicin-linezolid arm

For 8 weeks

DRUG	<40KG	40KG- 54KG	55KG - 70KG	≥71KG
Rifampicin	35mg/kg (rounded to nearest 150mg, maximum 2100mg)*			
Isoniazid	150mg	225mg	300mg	375mg
Pyrazinamide	800mg	1200mg	1600mg	2000mg
Ethambutol	550mg	825mg	1100mg	1375mg
Linezolid	600mg			

*The dose of rifampicin was decreased from 35mg/kg to 20mg/kg for all participants from 1st November 2019 after 88 participants had been enrolled in this arm (recommendation of the Trial Steering Committee as a precaution following a death from drug-induced liver injury in this arm).

C. Rifampicin-clofazimine arm

For 8 weeks

DRUG	<40KG	40KG- 54KG	55KG - 70KG	≥71KG
Rifampicin	35mg/kg (rounded to nearest 150mg, maximum 2100mg)			
Isoniazid	150mg	225mg	300mg	375mg
Pyrazinamide	800mg	1200mg	1600mg	2000mg
Ethambutol	550mg	825mg	1100mg	1375mg
Clofazimine	200mg			

D. Rifapentine-linezolid arm

For 8 weeks

DRUG	<40KG	40KG- 54KG	55KG - 70KG	≥71KG
Isoniazid	5mg/kg rounded to nearest 100mg		300mg	
Pyrazinamide	25mg/kg rounded to nearest 500mg	1000mg	1500mg	2000mg
Rifapentine	1200mg			
Linezolid	600mg			
Levofloxacin	1000mg			

E. Bedaquiline-linezolid arm

For 8 weeks

DRUG	<40KG	40KG- 54KG	55KG - 70KG	≥71KG
Bedaquiline	400 mg once daily for 2 weeks then 200mg three times a week			
Isoniazid	5mg/kg rounded to nearest 100mg		300mg	
Pyrazinamide	25mg/kg rounded to nearest 500mg	1000mg	1500mg	2000mg
Ethambutol	15mg/kg (rounded to nearest 100mg, maximum 1600mg)			
Linezolid	600mg			

OUTCOMES

Primary endpoint

- composite of death before week 96 or
- ongoing tuberculosis treatment or
- active tuberculosis at week 96.
- These outcomes were not considered to be primary-outcome events if retreatment had been completed and the participant did not have active disease at week 96.

Secondary endpoints

1) total treatment time
2) grade 3 or 4 adverse events
3) acquired drug resistance

STATISTICAL ANALYSIS

- For the test of noninferiority for the primary outcome with a margin of **12 percentage points** at a **one-sided level of 0.0125**, at least 180 participants were required to provide a **power of at least 85%**
- **Noninferiority of a treatment strategy** could be concluded if the upper limit of the two-sided **97.5% confidence interval** difference between the strategy group and the standard-treatment group was **less than 12 percentage points**
- Formal hypothesis testing was performed in **only the two strategy groups** with complete enrollment.
- Intention-to-treat population





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RESULTS

Table 2. Primary Efficacy Outcome.*

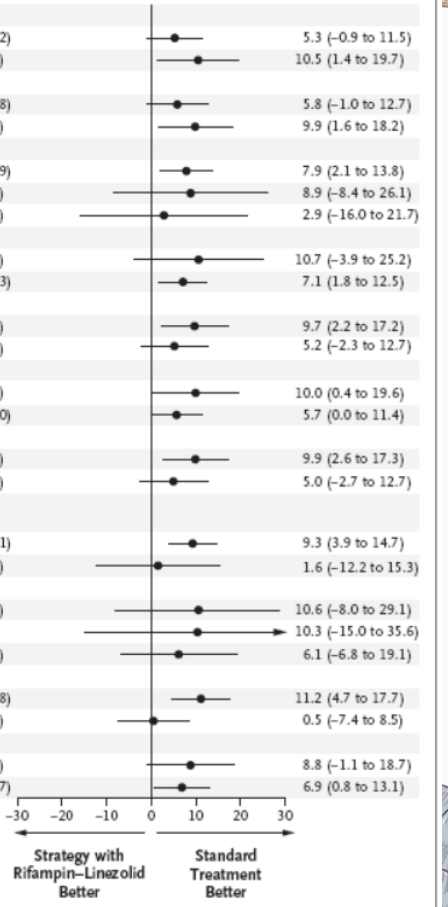
Outcome	Standard Treatment (N=181)	Strategy with Rifampin–Linezolid (N=184)	Strategy with Rifampin–Linezolid vs. Standard Treatment	Strategy with Bedaquiline–Linezolid (N=189)	Strategy with Bedaquiline–Linezolid vs. Standard Treatment
			Adjusted Difference (97.5% CI)†		Adjusted Difference (97.5% CI)†
Intention-to-treat population‡					
Primary outcome: composite of death, ongoing treatment, or active disease at wk 96 — no. (%)§	7 (3.9)	21 (11.4)	7.4 (1.7 to 13.2)	11 (5.8)	0.8 (–3.4 to 5.1)
Death before wk 96	2 (1.1)	5 (2.7)	—	1 (0.5)	—
Ongoing treatment at wk 96	2 (1.1)	8 (4.3)	—	5 (2.6)	—
Active disease at wk 96¶	1 (0.6)	4 (2.2)	—	3 (1.6)	—
Evaluation by telephone at wk 96 with no evidence of active disease but insufficient evidence of disease clearance when last seen	2 (1.1)	3 (1.6)	—	1 (0.5)	—
No evaluation at wk 96 and insufficient evidence of disease clearance when last seen	0	1 (0.5)	—	1 (0.5)	—
Outcomes classified as unassessable — no. (%)	1 (0.6)	1 (0.5)	—	2 (1.1)	—
Single positive culture at wk 96 but no other evidence of active disease‖	0	1 (0.5)	—	0	—
Death from a cause that was definitively unrelated to tuberculosis**	1 (0.6)	0	—	0	—
No evaluation at wk 96 and sufficient evidence of disease clearance when last seen	0	0	—	2 (1.1)	—
No primary outcome or outcome classified as unassessable — no. (%)	173 (95.6)	162 (88.0)	—	176 (93.1)	—
Assessable population††					
Primary outcome — no./total no. (%)	7/180 (3.9)	21/183 (11.5)	7.5 (1.7 to 13.2)	11/187 (5.9)	0.8 (–3.4 to 5.1)
Per-protocol population‡‡					
Primary outcome — no./total no. (%)	6/177 (3.4)	17/160 (10.6)	6.9 (0.9 to 12.8)	9/176 (5.1)	0.9 (–3.3 to 5.1)

Table 3. Secondary Outcomes.*

Outcome	Standard Treatment (N=181)	Strategy with Rifampin–Linezolid (N=184)	Strategy with Rifampin–Linezolid vs. Standard Treatment Difference (95% CI)†	Strategy with Bedaquiline–Linezolid (N=189)	Strategy with Bedaquiline–Linezolid vs. Standard Treatment Difference (95% CI)†
Participant-centered outcomes					
Total treatment time through wk 96 — days‡					
Total duration of treatment	180.2±37.9	105.7±80.1	−74.5 (−87.4 to −61.6)	84.8±65.3	−95.3 (−106.2 to −84.5)
Total qualifying treatment time	177.3±35.6	101.6±74.9	−75.7 (−87.7 to −63.6)	83.8±64.2	−93.5 (−104.0 to −82.9)
Acceptability scores§					
Difficulty score	1.5±1.7	2.4±2.2	1.0 (0.6 to 1.4)	1.8±2.0	0.4 (0.0 to 0.8)
Anxiety score	3.6±2.2	3.9±2.0	0.3 (−0.1 to 0.8)	3.4±2.0	−0.2 (−0.6 to 0.3)
Motivation score	6.2±3.9	8.0±3.0	1.8 (1.1 to 2.5)	8.1±2.9	1.9 (1.2 to 2.6)
Preferred treatment recommendation to others — no. (%)					
2-Mo treatment	NA	126/176 (71.6)	—	141/180 (78.3)	—
6-Mo treatment	NA	35/176 (19.9)	—	25/180 (13.9)	—
No preference	NA	15/176 (8.5)	—	14/180 (7.8)	—
Quality-of-life scores¶					
Mental health summary score	57.5±0.5	57.5±0.5	−0.02 (−1.35 to 1.30)	57.8±0.5	0.33 (−1.02 to 1.68)
Physical health summary score	56.7±0.5	56.8±0.5	0.06 (−1.24 to 1.37)	56.7±0.5	0.00 (−1.25 to 1.26)
Health-status score‖	0.99±0.0	0.98±0.1	−0.01 (−0.02 to 0.00)	0.98±0.1	−0.01 (−0.02 to 0.01)
Body weight**					
Change from baseline — kg	5.8±4.8	5.6±4.7	−0.3 (−1.3 to 0.6)	6.1±4.8	0.2 (−0.7 to 1.2)
Change from baseline — %	11.9±10.0	11.4±9.8	−0.8 (−2.8 to 1.3)	12.1±9.8	0.3 (−1.7 to 2.4)
Safety outcomes					
Adverse events through wk 96 — no. (%)					
Any grade 3 or 4 adverse event	29 (16.0)	32 (17.4)	1.4 (−6.4 to 9.2)	30 (15.9)	−0.2 (−7.9 to 7.4)
Any serious adverse event	11 (6.1)	18 (9.8)	3.7 (−2.1 to 9.7)	14 (7.4)	1.3 (−4.2 to 6.9)
Death††	3 (1.7)	5 (2.7)	1.1 (−2.4 to 4.8)	1 (0.5)	−1.1 (−4.3 to 1.5)
Respiratory disability at wk 96 — no. (%)‡‡					
Grade on MRC breathlessness scale ≥3	0	2.7 (1.5)	1.5 (−0.5 to 3.5)	2.7 (1.4)	1.4 (−0.5 to 3.3)
FEV_1 <50% of predicted value	24.3 (13.4)	20.5 (11.1)	−1.1 (−8.7 to 6.4)	22.4 (11.8)	0.1 (−7.8 to 7.9)

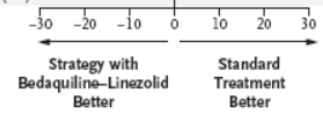
A Primary Outcome in Strategy Group with Initial Rifampin–Linezolid Regimen vs. Standard-Treatment Group

Subgroup	Strategy with Rifampin–Linezolid *no. of participants with event/total no. (%)*	Standard Treatment *no. of participants with event/total no. (%)*	Difference (95% CI) *percentage points*
Sex			
Male	11/113 (9.7)	5/119 (4.2)	5.3 (–0.9 to 11.5)
Female	10/71 (14.1)	2/62 (3.2)	10.5 (1.4 to 19.7)
Age			
18–34 yr	11/109 (10.1)	5/104 (4.8)	5.8 (–1.0 to 12.7)
35–65 yr	10/75 (13.3)	2/77 (2.6)	9.9 (1.6 to 18.2)
Country			
Indonesia or Philippines	15/139 (10.8)	4/139 (2.9)	7.9 (2.1 to 13.8)
Uganda	4/25 (16.0)	2/28 (7.1)	8.9 (–8.4 to 26.1)
India or Thailand	2/20 (10.0)	1/14 (7.1)	2.9 (–16.0 to 21.7)
Education			
0–7 yr	7/37 (18.9)	4/49 (8.2)	10.7 (–3.9 to 25.2)
≥8 yr	14/147 (9.5)	3/132 (2.3)	7.1 (1.8 to 12.5)
Smoking status			
Never	11/88 (12.5)	2/89 (2.2)	9.7 (2.2 to 17.2)
Current or former	10/96 (10.4)	5/92 (5.4)	5.2 (–2.3 to 12.7)
Body-mass index			
<18.5	13/80 (16.3)	5/79 (6.3)	10.0 (0.4 to 19.6)
≥18.5	8/104 (7.7)	2/102 (2.0)	5.7 (0.0 to 11.4)
Cavitation on chest radiograph			
Absent	10/83 (12.0)	1/87 (1.1)	9.9 (2.6 to 17.3)
Present	11/101 (10.9)	6/94 (6.4)	5.0 (–2.7 to 12.7)
Proportion of lung affected on chest radiograph			
≤50%	17/149 (11.4)	3/140 (2.1)	9.3 (3.9 to 14.7)
>50%	4/35 (11.4)	4/41 (9.8)	1.6 (–12.2 to 15.3)
WHO smear grade			
Negative	7/57 (12.3)	1/46 (2.2)	10.6 (–8.0 to 29.1)
Scanty or 1+	7/76 (9.2)	0/65	10.3 (–15.0 to 35.6)
2+ or 3+	7/51 (13.7)	5/69 (7.2)	6.1 (–6.8 to 19.1)
Grade on MRC breathlessness scale			
Grade 1	17/126 (13.5)	3/106 (2.8)	11.2 (4.7 to 17.7)
Grade 2 or higher	4/58 (6.9)	4/75 (5.3)	0.5 (–7.4 to 8.5)
Relapse risk			
Low	7/57 (12.3)	2/47 (4.3)	8.8 (–1.1 to 18.7)
Intermediate or high	14/127 (11.0)	5/134 (3.7)	6.9 (0.8 to 13.1)



B Primary Outcome in Strategy Group with Initial Bedaquiline–Linezolid Regimen vs. Standard-Treatment Group

Subgroup	Strategy with Bedaquiline–Linezolid *no. of participants with event/total no. (%)*	Standard Treatment *no. of participants with event/total no. (%)*	Difference (95% CI) *percentage points*
Sex			
Male	5/116 (4.3)	5/119 (4.2)	0.1 (–4.4 to 4.6)
Female	6/73 (8.2)	2/62 (3.2)	3.6 (–3.9 to 11.2)
Age			
18–34 yr	4/95 (4.2)	5/104 (4.8)	–1.8 (–7.0 to 3.4)
35–65 yr	7/94 (7.4)	2/77 (2.6)	4.8 (–1.1 to 10.7)
Country			
Indonesia or Philippines	6/145 (4.1)	4/139 (2.9)	1.3 (–3.0 to 5.5)
Uganda	4/27 (14.8)	2/28 (7.1)	7.7 (–8.8 to 24.1)
India or Thailand	1/17 (5.9)	1/14 (7.1)	–1.3 (–18.8 to 16.3)
Education			
0–7 yr	3/48 (6.3)	4/49 (8.2)	–2.4 (–12.1 to 7.3)
≥8 yr	8/141 (5.7)	3/132 (2.3)	2.8 (–1.6 to 7.3)
Smoking status			
Never	8/107 (7.5)	2/89 (2.2)	3.7 (–2.2 to 9.5)
Current or former	3/82 (3.7)	5/92 (5.4)	–1.0 (–7.0 to 5.0)
Body-mass index			
<18.5	4/76 (5.3)	5/79 (6.3)	–1.3 (–8.5 to 6.0)
≥18.5	7/113 (6.2)	2/102 (2)	3.3 (–2.0 to 8.6)
Cavitation on chest radiograph			
Absent	5/81 (6.2)	1/87 (1.1)	3.4 (–2.4 to 9.2)
Present	6/108 (5.6)	6/94 (6.4)	–0.6 (–7.0 to 5.9)
Proportion of lung affected on chest radiograph			
≤50%	8/151 (5.3)	3/140 (2.1)	2.8 (–1.0 to 6.6)
>50%	3/38 (7.9)	4/41 (9.8)	–2.2 (–14.4 to 10.1)
WHO smear grade			
Negative	1/50 (2.0)	1/46 (2.2)	0.9 (–16.8 to 18.7)
Scanty or 1+	4/77 (5.2)	0/65	6.4 (–18.7 to 31.6)
2+ or 3+	6/62 (9.7)	5/69 (7.2)	2.0 (–9.8 to 13.8)
Grade on MRC breathlessness scale			
Grade 1	3/112 (2.7)	3/106 (2.8)	1.1 (–3.0 to 5.2)
Grade 2 or higher	8/77 (10.4)	4/75 (5.3)	3.5 (–5.0 to 11.9)
Relapse risk			
Low	1/50 (2.0)	2/47 (4.3)	–0.8 (–7.3 to 5.6)
Intermediate or high	10/139 (7.2)	5/134 (3.7)	2.6 (–2.6 to 7.8)



04

DISCUSSION

DISCUSSION

- The results of the TRUNCATE-TB trial showed that a strategy involving initial treatment with an 8-week regimen that contained bedaquiline and linezolid was noninferior to standard treatment with respect to the risk of a composite clinical outcome at week 96.
- This treatment strategy was associated with a shorter initial course and with a shorter total duration of treatment than was standard treatment.
- Treatment strategy reported a higher level of motivation to adhere to an 8-week initial course than to standard treatment.





DISCUSSION

- Future cost-effectiveness analyses with reduced treatment duration are under way.
- Bedaquiline may be well-suited for a 2-month initial course because its long half-life may extend efficacy beyond treatment completion, with a low risk of drug resistance.
- The results of this trial suggest that there may be value in considering a shift in tuberculosis management to a strategy involving initial treatment for the minimum of persons with tuberculosis.





STRENGTHS

- the pragmatic design, the use of outcome measures that are relevant to persons with tuberculosis and to treatment programs, and the inclusion of diverse treatment clinics in high-burden countries, mainly in Asia.
- The primary outcome was shown in Subgroup Analysis, that allows researchers to test hypothesis in more detail.

LIMITATIONS

- The open-label design
- Noninferiority of the treatment strategy was assessed with only two regimens, and the strategy could be refined with the use of alternative regimens.
- No HIV-positive persons were enrolled; further evaluation is warranted in this population

IMPLICATIONS

จากการศึกษาพบว่า 8-week regimen ที่มี Bedaquiline – Linezolid มีประสิทธิภาพทางคลินิกที่ไม่ด้อยไปกว่าสูตรมาตรฐาน IRZE (24-week regimen) แต่พบว่าผู้ป่วยให้ความร่วมมือในการใช้ยามากกว่าในกลุ่ม 8-week regimen นอกจากนี้ยังพบว่าการเกิด Adverse events ก็ไม่มีความต่างกันในสอง regimen นี้ ซึ่งแนวทางการรักษาแบบ short-course นี้อาจเป็นอีกหนึ่งแนวทางในการใช้รักษาผู้ป่วยวัณโรคชนิด Rifampin-Susceptible TB เพื่อเพิ่มความร่วมมือในการใช้ยา และลดโอกาสการเกิดวัณโรคดื้อยาต่อไปในอนาคต

E. Bedaquiline-linezolid arm

For 8 weeks

DRUG	<40KG	40KG- 54KG	55KG - 70KG	≥71KG
Bedaquiline	400 mg once daily for 2 weeks then 200mg three times a week			
Isoniazid	5mg/kg rounded to nearest 100mg		300mg	
Pyrazinamide	25mg/kg rounded to nearest 500mg	1000mg	1500mg	2000mg
Ethambutol	15mg/kg (rounded to nearest 100mg, maximum 1600mg)			
Linezolid	600mg			

THANKS

DOES ANYONE HAVE ANY QUESTIONS?







